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Problem

- Left-to-right pretraining prevents every layer from jointly conditioning on both left and right context.
- This is especially harmful for token-level tasks such as question answering, where future context matters.
- Feature-based transfer also requires substantial task-specific architectures; BERT seeks a unified fine-tuning recipe.

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Representation

ONE BIDIRECTIONAL ENCODER

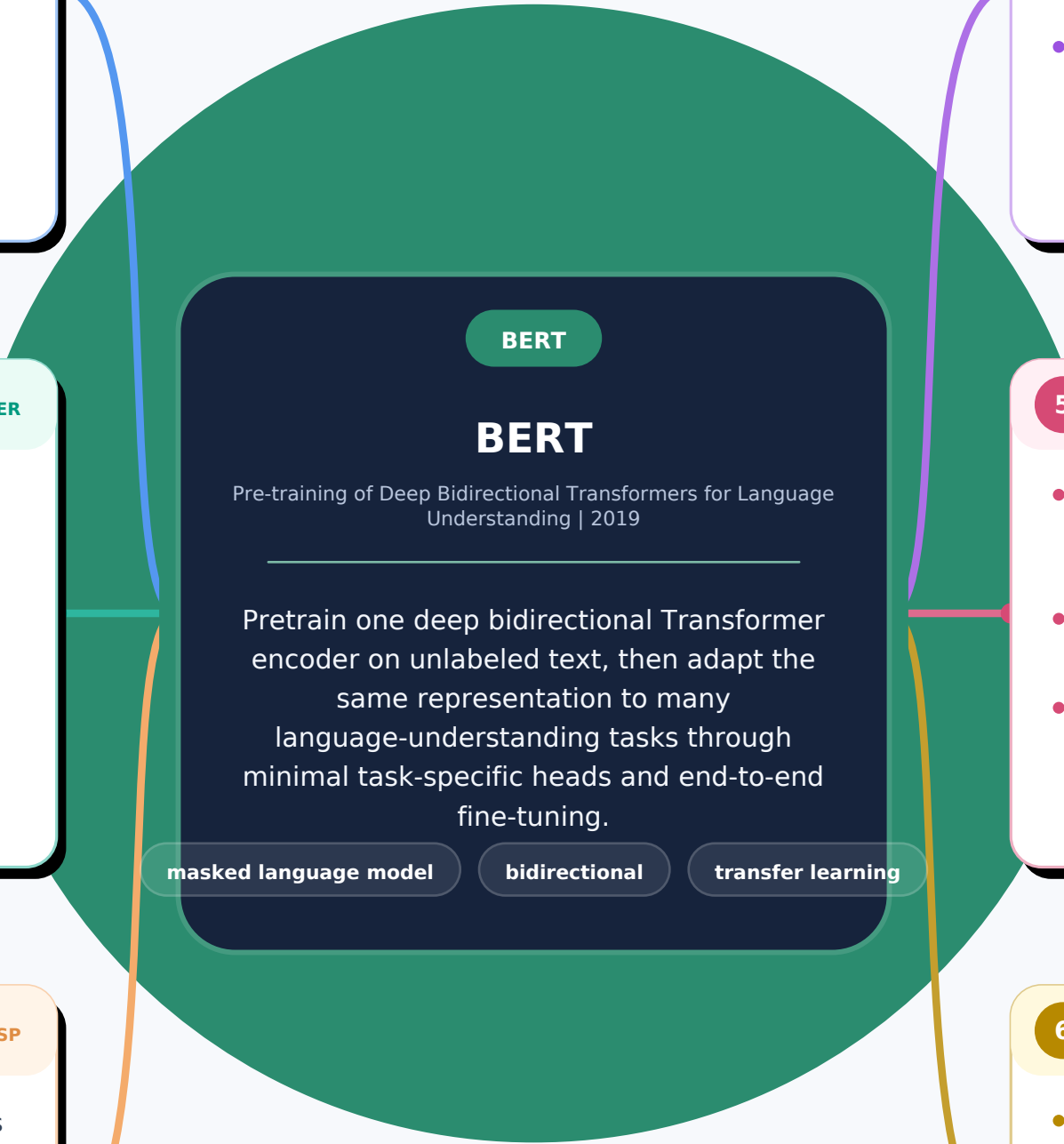
- BERTBASE: 12 layers, 768 hidden, 12 heads and 110M parameters; BERTLARGE: 24/1024/16 and 340M.
- Inputs sum WordPiece token, segment and position embeddings; [CLS] aggregates sequences and [SEP] separates text pairs.
- Bidirectional self-attention jointly encodes sentence pairs, replacing separate encoding plus cross-attention pipelines.

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Pre-training

MLM + NSP

- Mask 15% of WordPieces and predict only those originals from full context.
- Of selected tokens, 80% become [MASK], 10% random tokens and 10% stay unchanged to reduce train/test mismatch.
- Next Sentence Prediction classifies true versus random continuations with a 50/50 mixture to learn text-pair relations.



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Transfer

PRETRAIN ONCE, FINE-TUNE ALL

- Pretrain on BooksCorpus (800M words) plus English Wikipedia (2.5B words).
- For classification, add a softmax over [CLS]; for QA or tagging, predict from token vectors.
- All pretrained parameters are fine-tuned end to end, usually with only one new output layer and modest task compute.

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Evidence

ELEVEN TASKS, NEW SOTA

- BERTLARGE reaches 80.5 official GLUE and 86.7% MultiNLI accuracy, improving prior results by 7.7 and 4.6 points.
- SQuAD v1.1 reaches 93.2 test F1; SQuAD v2.0 reaches 83.1 test F1.
- Ablations show deep bidirectionality and NSP both help; larger pretrained models improve even small downstream datasets.

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Meaning & limits

- BERT made pretrain-then-fine-tune a simple default and reduced dependence on handcrafted task architectures.
- MLM predicts only a subset of tokens and introduces [MASK] mismatch; NSP was later revisited by follow-up work.
- Each task still stores a full fine-tuned checkpoint, motivating parameter-efficient methods such as LoRA.